

## EV CHARGING DEMAND PREDICTION

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## LEARNING OBJECTIVES

1. Understand the need for EV charging demand prediction in smart grid and transport systems.
2. Identify the key factors that influence EV charging demand, such as location, time, and weather.
3. Explain various machine learning and time-series methods use for demand forecasting.
4. Describe the data collection and preprocessing steps required for building predictive models.
5. Visualize feature importance to identify key drives influencing EV adaption
6. Evaluate model performance using accuracy, classification report and confusion matrix



## Tools and Technology used:

### DATA COLLECTION AND MANAGEMENT:

1. APIs (Google Maps, Weather APIs)
2. SQL/ NO SQL DATABASES ( MySQL, MongoDB)
3. CSV/JSON files for raw data storage

### DATA PROCESSING & ANALYSIS

1. PYTHON (Pandas , NumPy)
2. Jupyter notebooks for prototyping
3. Google COLAB

### MACHINE LEARNING & FORECASTING

1. Scikit –learn : regression , classification , feature selection
2. XGBoost / LightGBM: Boosted trees for high accuracy

### WEB APP:

STREAMLIT for interactive development

VERSION CONTROL: Git & GitHub for code management

## Methodology

### Step1: Data collection

Gather historical charging data from public and private EV stations

Include features such as : timestamp, charging duration , energy consumed, location, weather, traffic patterns.

Step2: Data preprocessing: handled missing values, encoded categorical data and created new features.

Step3: Exploratory Data Analysis(EDA): Explored data patterns and trends using visualizations with Matplotlib and Seaborn.

Step4: Model Training: Built a random forest classifier to predict EV adoption using selected features.

Step5: MODEL Evaluation: Evaluated performance using accuracy, confusion matrix and feature importance.

Step6: FORECASTING/ Deployment: Saved the trained model with Job lib and deployed an interactive web app using Streamlit.

## Problem Statement:

### EV Charging demand prediction in INDIA

#### RISING EV adoption

Government push via fame-II and National Electric Mobility Mission, TARGET: 30% of vehicles electric by 2030 , rapid growth in 2-wheelers , 3-wheelers , and commercial EV fleets

#### Challenges in Charging infrastructure

Uneven station distribution- urban vs rural gaps , grid limitations- especially in peak hours and tier-2/3 cities

#### Why prediction is critical in India ?

High population density , regional variation in vehicle types, usage patterns, and grid strength

Energy dependency on fossil fuels makes load prediction crucial for renewables integration

#### Localized prediction solutions

City wise demand forecasting (Delhi, Mumbai, Bengaluru)

Using of machine learning and IoT- based smart meters

Leverage traffic data, energy patterns, and weather forecasts for precision

## **Solution:**

### **National EV Data platform**

**Centralized portal to collect and share real time charging data across states, enable interoperability between different charging networks and apps.**

### **Smart Grid Integration**

**Integrate predictions with state DISCOMs and national grid, allow load balancing and demand response mechanisms , avoid overloading during peak charging times**

**Preprocessed and engineered features like lag values, rolling averages, and growth slopes**

**Trained multiple regression models : decision tree, SVR, and random forest.**

**Selected random forest regressor based on best performance**

**Forecasted EV adoption for the next 36 months at the country level**

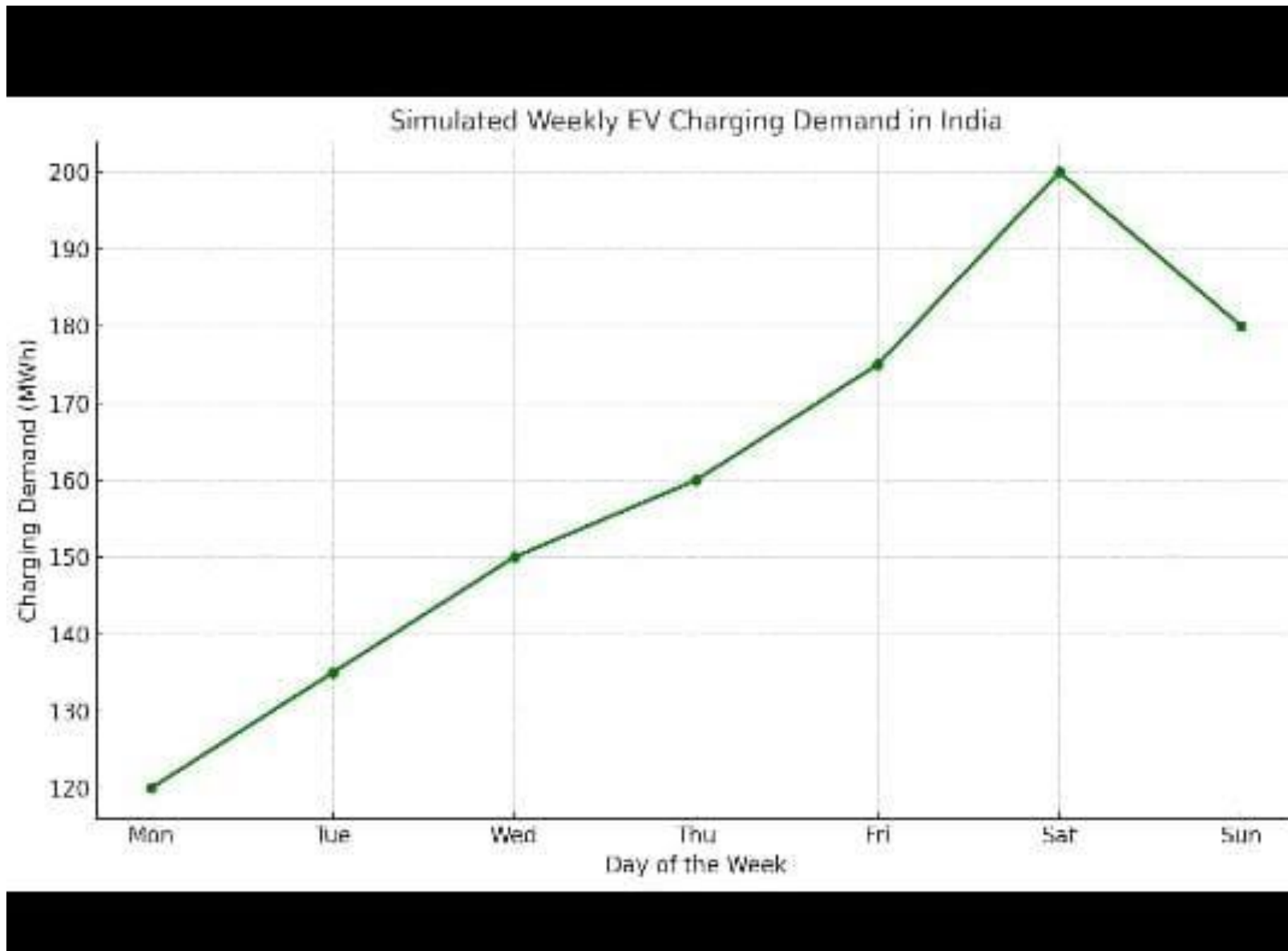
**Deployed the solution using an interactive streamlit web app with forecasting and comparison features.**

**Github repository link : <https://github.com/simpal4765>**



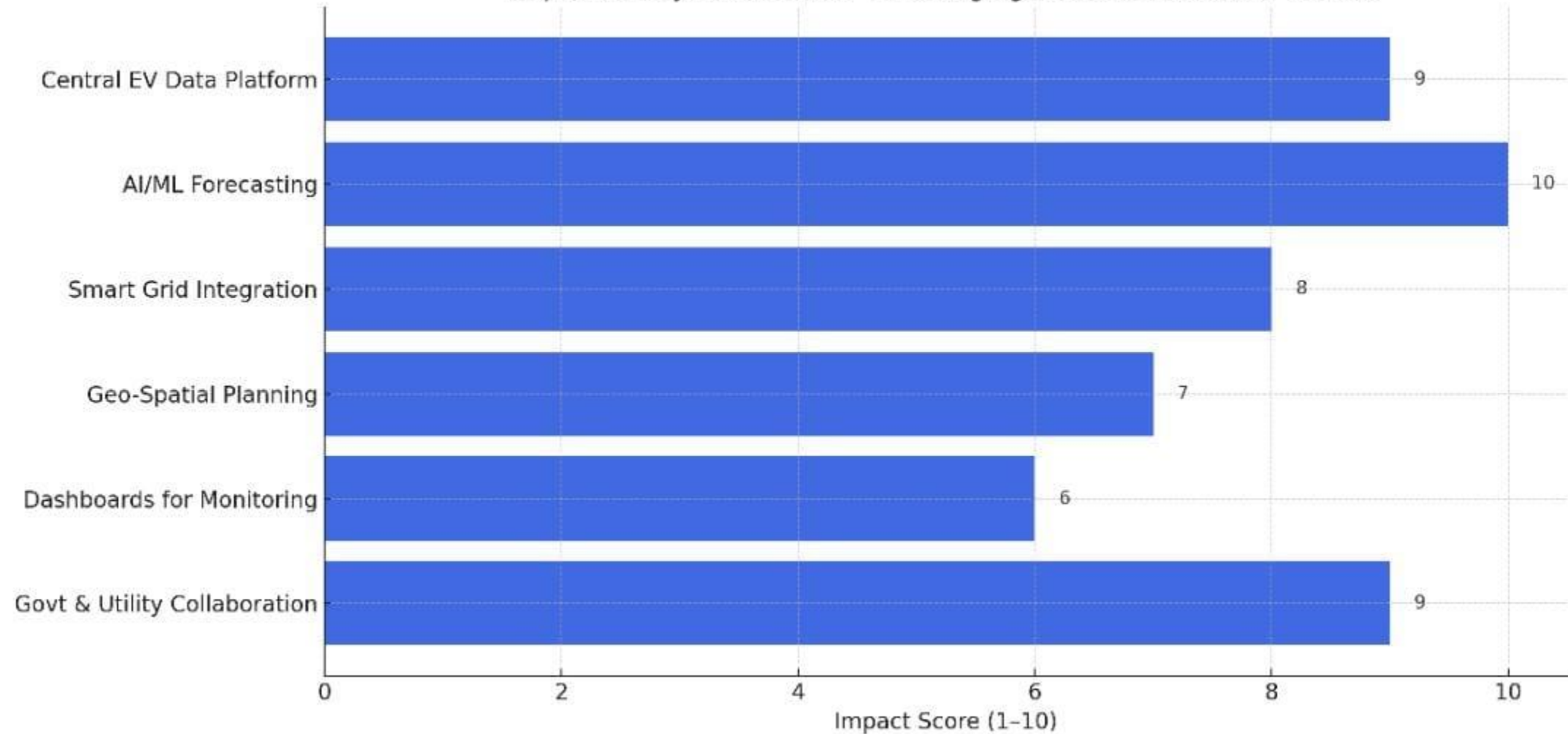
## Screenshot of Output:







Impact of Key Solutions for EV Charging Demand Prediction in India



## Conclusion:



Slide: Conclusion



India's EV growth requires a smart and scalable charging ecosystem.



Predicting demand accurately ensures efficient energy use, reduces congestion, and enhances user satisfaction.



AI/ML-based models, when integrated with real-time data and grid systems, can provide high-accuracy forecasting.



State-specific strategies and national coordination are both essential for balanced infrastructure development.



Ultimately, predictive systems will play a key role in supporting India's transition to clean, connected, and electric mobility.---